

Documentary Film of Low Vision Boxing: Collaborative Design of Adaptive Kickboxing Experiences

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Figure 1: Full-body view of kickboxers wearing the developed prototype, trying out the prototype.

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ABSTRACT

In this video showcase, we address the challenge of physical in-activity among individuals with low vision by introducing adaptive kickboxing—a newly conceptualized sport that employs assistive technology tailored to their needs. Through collaborative design with a low vision participant, we develop equipment like gloves and shin guards that amplify the perceptibility of movements in kickboxing, enabling engagement and safety in the sport. Our research extends beyond facilitating exercise to creating opportunities for competitive aspirations within a sport transformed for inclusivity.

KEYWORDS

low vision, adaptive kickboxing, collaborative design, sports accessibility

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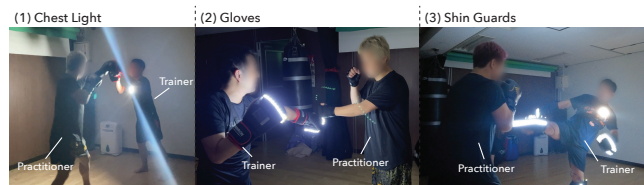


Figure 2: Three key methods of visual information presentation: (1) chest light, (2) gloves, and (3) shin guards. The photos in the figure capture the moments, showcasing the practitioner's interaction with each newly developed prototype.

1 LOW VISION BOXING

In this video showcase, we elaborate on the development of assistive kickboxing equipment for individuals with low vision. The

design process was a collaborative effort between researchers, a visually impaired kickboxing practitioner, and an expert kickboxing trainer. The insights from these contributors informed prototype developments to address specific visual challenges in kickboxing.

The key challenges identified included ascertaining the opponent's location, recognizing arm movement, and detecting leg movement. Solutions involved integrating visual aids using light sources that allowed for better visibility and orientation. The prototypes included enhanced equipment with lights (Figure 2):

- A compact LED on the chest to indicate the opponent's position.
- Light tape on the gloves to signal arm movement.
- Similarly, light tape on shin guards to highlight leg movements.

The iterative design process with continuous feedback ensured that the enhancements were practical and supportive of the needs of low-vision kickboxers. The final prototypes provided necessary visual cues without hindering movement or comfort, catered specifically to the distinct requirements of kickboxing participants with low vision.

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